PETER CHRISTIAN AGBENYEGA

Cloud Security & DevSecOps Engineer | AWS Security Specialist | Infrastructure Architect

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AWS SAA-C03 | AWS SAP-C02 (DevOps Professional) | AWS SCS-C02 (Security Specialty)

PROPOSED ENDEAVOR - NATIONAL INTEREST

Peter Agbenyega's proposed endeavor is the design, publication, and deployment of secure cloud infrastructure reference architectures that directly reduce cloud misconfiguration - identified by CISA and IBM as a leading, preventable cause of US data breaches, costing American organizations an average of $4.45 million per incident. Through open-source reference implementations, an active consulting practice, and graduate research in cloud computing systems, Peter advances the national security and economic resilience of American cloud infrastructure. His work is publicly available, freely reusable, and directly applicable to US small and medium-sized businesses, healthcare organizations, and critical infrastructure operators that lack the resources to maintain dedicated security teams.

AWS CERTIFICATIONS

AWS Security - Specialty
SCS-C02
IAM, KMS, GuardDuty, CloudTrail, VPC Security
Most relevant to endeavor

AWS DevOps Engineer - Professional
SAP-C02
CI/CD, IaC, Automation, Monitoring
Professional level

AWS Solutions Architect - Associate
SAA-C03
VPC, EC2, S3, IAM, Route53, RDS
Foundation architecture

EDUCATION

Master of Science - Cloud Computing Systems | University of Maryland Global Campus — **In Progress - 2024 to Present**
Adelphi, Maryland | Accredited US Graduate Institution

- Active coursework in cloud architecture, distributed systems, and advanced cloud security
- Summer 2025 courses: CLCS 605, CLCS 615 | Planned fall: CLCS 625, 635, 645
- Graduate research directly informs open-source cloud security reference architecture work

Graduate Certificate - Cybersecurity Technology | University of Maryland Global Campus — **In Progress - 2024 to Present**
Concurrent with MS | Focused on cloud-native security and compliance frameworks

Bachelor of Education - Mathematics | University of Cape Coast — **Graduated June 2012**
Cape Coast, Ghana | Analytical foundation applied to infrastructure design and security modeling

OPEN-SOURCE SECURITY REFERENCE ARCHITECTURES - PUBLIC CONTRIBUTIONS

The following reference architectures are published under open-source licenses on GitHub and are freely available to any US organization seeking to strengthen its cloud security posture. Each represents a direct, verifiable contribution to the national interest by reducing cloud misconfiguration risk.

ResilientOps AWS Reference Architecture | github.com/Peter-Agbenyega/ResilientOps-Cloud-Architecture — **Public - 2025**

- Production-grade AWS infrastructure implementing multi-tier networking, application load balancing, and auto-scaling patterns essential to American enterprise resilience
- Terraform modules cover: VPC design, security groups, ALB, Auto Scaling Groups, multi-AZ deployment
- Directly addresses NIST SP 800-144 guidelines for cloud infrastructure security

ClusterShield360 - EKS Security Architecture | github.com/Peter-Agbenyega/ClusterShield360 — **Public - 2025**

- Secure Kubernetes infrastructure on Amazon EKS with RBAC, network policies, and IAM role separation
- Security-focused container orchestration patterns for organizations running containerized workloads at scale
- Addresses CIS Kubernetes Benchmark controls for US regulated industry deployments

CloudShield360 - AWS Secure Landing Zone | github.com/Peter-Agbenyega/cloudshield360 — **Public - 2025**

- Modular production-ready AWS secure landing zone implementing Infrastructure-as-Code baseline patterns
- Provides US organizations a controlled, compliant foundation for cloud adoption from day one
- Implements AWS Well-Architected Framework Security Pillar requirements across all deployed resources

DoD-Level CI/CD Security Pipeline | github.com/Peter-Agbenyega/Dod-Level-CICD-Project — **Public - 2025**

- Department of Defense-inspired security pipeline integrating Gitleaks, Checkov, Trivy, SonarCloud, and OWASP ZAP

- Multi-stage security gate architecture prevents deployment of misconfigured or vulnerable infrastructure
- Freely reusable by any US organization seeking defense-grade deployment security practices

secure-cloud-baseline-architecture | github.com/Peter-Agbenyega/secure-cloud-baseline-architecture **Public - 2024**

- Non-commercial cloud security reference architecture focused on secure-by-design AWS baseline patterns
- Published explicitly for public-interest cybersecurity resilience - no commercial restriction

MultiZoneX360 - High-Availability Architecture | github.com/Peter-Agbenyega/MultiZoneX360 **Public - 2025**

- Multi-zone AWS architecture demonstrating resilient infrastructure patterns for business continuity
- Applicable to US healthcare, finance, and critical infrastructure requiring near-zero downtime

PROFESSIONAL EXPERIENCE

Founder & Principal Cloud Security Engineer | Cloud Nexus Hub LLC **2024 to Present**

Utica, New York | Registered US LLC | cloudnexus360.com

- Founded Cloud Nexus Hub LLC to deliver enterprise-grade cloud security consulting to US small and medium-sized businesses that cannot afford dedicated security engineering teams
- Designed and deployed secure AWS infrastructure for client Adcar Technology LLC - VPC, EC2, IAM, S3 security, Auto Scaling - following AWS Well-Architected Framework security pillar
- Built and launched Cloud Nexus Pilot (cloudnexuspilot.com) - an AI-powered interview preparation platform for cloud and DevSecOps professionals, featuring AWS Security Specialty practice questions drawn from SCS-C02 expertise
- Built and launched ubagstore.com - a full e-commerce platform serving Uncle Bakarr African Grocery, Utica, NY - providing digital commerce infrastructure for an underserved, minority-owned small business
- Built and launched womiex.com - a bilingual (EN/FR) international trade platform for WOMIEX LLC, connecting DRC mineral producers with US industrial buyers and supporting ethical US supply chain security
- Published 6 open-source AWS security reference architectures freely available on GitHub for US organizational use

Cloud Solutions Architect | DevSecOps Consultant | Adcar Technology LLC - Remote **April 2024 to Present**

Client engagement through Cloud Nexus Hub LLC

- Designed, deployed, and secured AWS cloud architectures supporting enterprise workloads using EC2, Lambda, VPC, IAM, and CloudTrail
- Reduced infrastructure provisioning time by 70% through Terraform and CloudFormation automation
- Developed secure CI/CD pipelines using GitHub Actions and Docker aligned with SOC2 and HIPAA compliance requirements
- Integrated AWS GuardDuty, WAF, KMS, and Config - delivering continuous security monitoring and auto-remediation
- Implemented policy-as-code using Rego (OPA) to enforce compliance guardrails at scale across AWS environments
- Deployed Kubernetes workloads on Amazon EKS with Helm charts, RBAC, and automated security scanning

Cloud Engineer | DevOps Engineer | Adkar Technology LLC - Remote **July 2019 to March 2024**

Accra, Ghana and Remote

- Led development of enterprise-grade, cloud-native platforms on AWS, applying security-first design principles across all architecture layers
- Built CI/CD pipelines integrating Jenkins, Git, and Docker - reducing time-to-deployment while strengthening security gate enforcement
- Conducted vulnerability assessments and hardened IAM roles using a least-privilege model - reducing the AWS attack surface across client accounts
- Managed PostgreSQL, MongoDB, and DynamoDB for scalable and compliant application backends
- Led migration of monolithic systems to microservices on Amazon EKS - improving deployment velocity and security isolation

TECHNICAL EXPERTISE

Cloud Platforms	AWS (EC2, VPC, Lambda, RDS, S3, IAM, CloudTrail, GuardDuty, KMS, WAF, Config, Security Hub, Route53)
Infrastructure as Code	Terraform, AWS CloudFormation, OPA Rego, Ansible - automated provisioning and compliance
Security Tools	Gitleaks, Checkov, Trivy, OWASP ZAP, SonarCloud, AWS GuardDuty, IAM Policy Analyzer
DevSecOps Pipeline	GitHub Actions, Jenkins, AWS CodePipeline - multi-stage security gate CI/CD implementation
Containers	Docker, Kubernetes (EKS), Helm Charts, ArgoCD - secure container orchestration
Monitoring	AWS CloudWatch, ELK Stack, Datadog - infrastructure observability and security alerting

Compliance Frameworks	NIST CSF, CIS AWS Benchmarks, AWS Well-Architected, SOC2, HIPAA
Languages	Python, Bash, Rego (OPA), SQL, TypeScript - infrastructure automation and security tooling
Full-Stack Development	Next.js, React, FastAPI, PostgreSQL, SQLite - SaaS product development for Cloud Nexus Hub

NATIONAL INTEREST CONTRIBUTION SUMMARY

Substantial Merit

AWS Security Specialty - one of AWS's most rigorous credentials. 3 certifications total including Professional level. MS Cloud Computing Systems in progress at accredited US institution.

National Importance

Cloud misconfiguration is a top CISA-cited cause of US breaches. Reference architectures address NIST CSF. Work protects US SMBs - 99.9% of all American businesses. DoD-level security pipeline implementation.

Well Positioned

5+ years hands-on cloud security experience. 6 live open-source reference architectures. Active US-based LLC with real clients. Community impact through digital infrastructure for minority-owned businesses.

Portfolio: petercagbenyega.com | GitHub: github.com/Peter-Agbenyega | Company: cloudnexus360.com

References and supporting documentation available upon request.